

Supplementary Information for “Harmonicity accounts for the tritone’s shortfall on the dissonance curve for dyads”

Daniel Chen

Independent Researcher, Taipei, Taiwan

boenchen1112@gmail.com ORCID 0009-0009-5840-5012

Section, table and figure numbers prefixed with S refer to this document; others refer to the main text. Every analysis is labelled pre-specified or exploratory where it is introduced; all analyses in Sections S4–S9 are exploratory unless stated otherwise. Units are within-listener SD unless stated; 95% CIs are in square brackets.

S1 Pooling: DerSimonian–Laird and Hartung–Knapp intervals

The main text pools experiments by random effects with the DerSimonian–Laird estimate of τ^2 and a Hartung–Knapp interval. Table S1 gives both intervals for the rows of main-text Table 4. Of 326 random-effects estimates in the released results, 22 change whether their interval excludes zero when the Hartung–Knapp interval replaces the DerSimonian–Laird one; the list is in `results/re_intervals.json`. Table S2 gives the dip, observed and left unexplained, for every experiment, with each model calibrated on the experiment itself.

Table S1: Random-effects pooled estimates with the DerSimonian–Laird (DL, normal) and Hartung–Knapp (HK, t on $k - 1$ df) 95% intervals, SD units, each model calibrated on the experiment itself. Harmonic: six US experiments with harmonic spectra; eight: adds pure tones and the bonang.

	Gap, harmonic			Gap, eight			Dip, harmonic		
	Est.	DL	HK	Est.	DL	HK	Est.	DL	HK
Observed	+0.26	[+0.18, +0.34]	[+0.16, +0.36]	+0.27	[+0.20, +0.33]	[+0.18, +0.35]	+0.30	[+0.24, +0.36]	[+0.22, +0.37]
After roughness (H&K)	+0.26	[+0.17, +0.35]	[+0.15, +0.37]	+0.26	[+0.19, +0.32]	[+0.18, +0.33]	+0.25	[+0.16, +0.33]	[+0.13, +0.36]
After roughness (Sethares)	+0.25	[+0.16, +0.35]	[+0.13, +0.38]	+0.25	[+0.17, +0.33]	[+0.16, +0.34]	+0.28	[+0.20, +0.36]	[+0.17, +0.39]
After harmonicity alone	+0.11	[−0.01, +0.24]	[−0.05, +0.27]	+0.14	[+0.03, +0.25]	[+0.02, +0.27]	+0.03	[−0.07, +0.13]	[−0.10, +0.16]
After roughness + harmonicity	+0.19	[+0.06, +0.32]	[+0.02, +0.36]	+0.20	[+0.10, +0.31]	[+0.08, +0.32]	+0.14	[+0.02, +0.25]	[−0.02, +0.29]
harmonicity template $i^{-0.75}$ (incon)	+0.20	[+0.08, +0.32]	[+0.05, +0.36]	+0.21	[+0.12, +0.31]	[+0.10, +0.32]	+0.15	[+0.04, +0.26]	[0.00, +0.30]
After revised H&K + harmonicity	+0.28	[+0.15, +0.40]	[+0.11, +0.44]	+0.27	[+0.17, +0.36]	[+0.15, +0.38]	+0.20	[+0.07, +0.34]	[+0.02, +0.39]
After Marjeh et al. composite ^a	+0.17	[+0.06, +0.28]	[+0.03, +0.31]	+0.18	[+0.09, +0.27]	[+0.08, +0.28]	+0.05	[−0.08, +0.19]	[−0.13, +0.23]

S2 The screen: rows examined afterwards

Table S3 lists the models examined after the pre-specified screen of main-text Section 5.4, in the format of main-text Table 6 (exploratory). The first row extends the tolerance grid beyond the pre-specified values; the held-out R^2 changes little over a broad range (main-text Figure 3), so the choice of 20 cents is representative rather than an optimum. The other rows use the timbre-relative harmonicity of Section S5, alone or added to the 20-cent term.

Table S2: Mean of the fourth and fifth minus the tritone (the dip), observed and left unexplained [95% participant-bootstrap CI], SD units, each model calibrated on the experiment itself. Bold: CI excludes zero. The composite of Marjeh et al. was fitted to these data.

Experiment	N	Observed	Left unexplained after		
			Roughness	Roughness + harmonicity	Composite
<i>United States</i>					
Harmonic, 3dB/oct	160	+0.38 [+0.25 , +0.51]	+0.29 [+0.17 , +0.42]	+0.10 [−0.02, +0.22]	+0.15 [+0.03 , +0.27]
5 equal harmonics	147	+0.27 [+0.14 , +0.40]	+0.12 [−0.02, +0.24]	−0.09 [−0.21, +0.04]	−0.13 [−0.26 , −0.01]
5 harmonics, no 3rd	151	+0.23 [+0.14 , +0.33]	+0.28 [+0.19 , +0.38]	+0.23 [+0.14 , +0.32]	+0.21 [+0.12 , +0.31]
Flute	188	+0.23 [+0.11 , +0.34]	+0.10 [−0.02, +0.22]	+0.03 [−0.09, +0.15]	−0.19 [−0.31 , −0.08]
Guitar	186	+0.33 [+0.20 , +0.46]	+0.32 [+0.19 , +0.45]	+0.25 [+0.12 , +0.39]	+0.14 [+0.01 , +0.27]
Piano	191	+0.39 [+0.27 , +0.51]	+0.36 [+0.24 , +0.48]	+0.28 [+0.17 , +0.41]	+0.13 [+0.01 , +0.25]
Pure tones	151	+0.28 [+0.17 , +0.39]	+0.28 [+0.17 , +0.39]	+0.20 [+0.10 , +0.32]	+0.10 [−0.01, +0.22]
Bonang (harmonic bass)	146	+0.36 [+0.23 , +0.49]	+0.30 [+0.18 , +0.42]	+0.04 [−0.09, +0.17]	+0.13 [+0.01 , +0.25]
Stretched (octave 2.1)	153	−0.03 [−0.15, +0.09]	−0.03 [−0.15, +0.08]	−0.03 [−0.15, +0.08]	−0.04 [−0.15, +0.08]
Compressed (octave 1.9)	154	+0.03 [−0.09, +0.14]	+0.03 [−0.09, +0.14]	+0.03 [−0.08, +0.15]	+0.02 [−0.09, +0.13]
<i>South Korea</i>					
Harmonic, 3dB/oct	40	+0.25 [+0.10 , +0.42]	+0.19 [+0.05 , +0.35]	−0.04 [−0.17, +0.09]	+0.07 [−0.08, +0.21]
Stretched (octave 2.1)	37	−0.06 [−0.21, +0.10]	−0.06 [−0.21, +0.10]	−0.06 [−0.21, +0.10]	−0.06 [−0.21, +0.10]
Compressed (octave 1.9)	45	−0.01 [−0.15, +0.15]	0.00 [−0.15, +0.16]	+0.04 [−0.11, +0.20]	0.00 [−0.14, +0.16]

Table S3: Models examined after the screen (exploratory; leave one US experiment out; point estimates). Columns as in main-text Table 6.

Model	Held-out R^2					Stretched, dip left	
	10 US	6 harm.	Gap left	Dip left	TT − other	at 6.00	at 6.42
H&K + harmonicity, σ grid extended to 60 cents	0.542	0.657	+0.19	+0.14	+0.16	−0.03	+0.09
H&K + timbre-relative harmonicity	0.515	0.640	+0.20	+0.19	+0.18	−0.06	+0.02
H&K + harmonicity(20) + timbre-relative(20)	0.534	0.644	+0.21	+0.14	+0.16	−0.02	+0.10

S3 The dip at every interval

Figure S1 repeats the dip analysis at every integer interval, pooled over the eight US experiments without stretched or compressed partials (fixed-effect, inverse-variance weights; harmonicity at the published 6.83-cent tolerance). This repeats an earlier analysis by the author, retained for comparison with the random-effects residual profile of main-text Table 7. Observed dips are largest at the minor second (+0.50) and the minor ninth (+0.41); the tritone ranks 4 of 14 (+0.30). After calibrated roughness the tritone has the largest unexplained dip (+0.26; next, the minor ninth, +0.19). After roughness and harmonicity it ranks 2 (+0.15, behind the major second at +0.19). A dip compares an interval with its two neighbours, so a large dip at the tritone after harmonicity can come from the fourth and fifth rated above the curve rather than from the tritone rated below it; the residual profile of the main text shows that it does.

S4 The in-tune bonus and the step

All analyses in this section are exploratory; the patterns were noticed in the residual profile. Three curves are compared: M1, H&K roughness plus harmonicity at 20 cents; M2, M1 plus a linear term in interval size (the curve used in the main text); M3, M1 plus the sharpness proxy of Section S8. For each, the held-out residual $r(c)$ is split at each equal-tempered interval c into a local bonus $b(c) = r(c) - [r(c - 0.5) + r(c + 0.5)]/2$ and a baseline $q(c) = [r(c - 0.5) + r(c + 0.5)]/2$, so that $r(c) = b(c) + q(c)$ and the gap $r(8) - r(6)$ splits into a bonus part $b(8) - b(6)$ and a baseline part $q(8) - q(6)$. The step is the coefficient of an indicator for $c \geq 8$ in a regression of $q(c)$ on c over 1–11 semitones (or 1–14), or the increment $q(8) - q(7)$ minus the mean of the other increments.

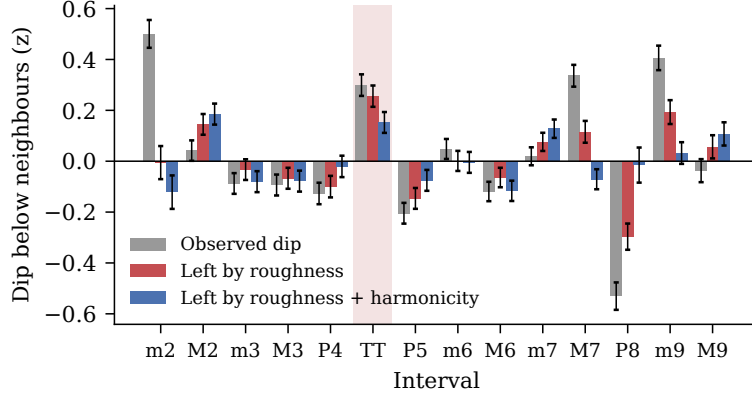


Figure S1: Dip of each integer interval below the mean of its two semitone neighbours, pooled over eight US experiments with fixed-effect (inverse-variance) weights; bars, ± 1.96 pooled SE. Grey: observed. Red: left by calibrated H&K roughness. Blue: left by roughness plus harmonicity at the published 6.83-cent tolerance. Each interval’s calibration excludes its own window. Shaded: tritone.

Plain means and random effects. Table S4 gives every quantity under both a plain mean over the six harmonic spectra, with a participant-bootstrap CI that treats the spectra as fixed, and random-effects pooling with a Hartung–Knapp CI, which treats them as a sample. The two disagree. Under plain means the bonus part (M2: $+0.07$ [$+0.01$, $+0.13$]), the baseline part ($+0.15$ [$+0.11$, $+0.19$]), the step ($+0.10$ [$+0.06$, $+0.14$]) and the tritone’s bonus shortfall relative to the other intervals ($+0.07$ [$+0.03$, $+0.11$]) all exclude zero, as they do under M1 and M3. Under random effects the bonus part ($+0.06$ [-0.03 , $+0.16$]), the step as an indicator ($+0.10$ [-0.03 , $+0.23$]) and the tritone’s shortfall ($+0.07$ [-0.01 , $+0.14$]; M1 $+0.07$ [0.00 , $+0.14$], M3 $+0.07$ [0.00 , $+0.14$]) include zero, while the gap ($+0.21$ [$+0.06$, $+0.36$]), the baseline part ($+0.15$ [$+0.02$, $+0.28$]) and the step as an increment ($+0.10$ [$+0.01$, $+0.19$]) do not. The main text reports the random-effects version, as for the primary estimand, because the question is about spectra in general. The plain-mean figures answer a narrower question, about these six spectra, and the statement in an earlier analysis by the author that the tritone receives about half the bonus rested on them; it is withdrawn. The same plain-mean contrast computed on the curve with the sharpness term (the version reported in that earlier analysis) is $+0.07$ [$+0.02$, $+0.11$].

One contrast holds under both: the major second and minor seventh receive a smaller bonus than the other eight intervals from the minor second to the major seventh (random effects -0.12 [-0.14 , -0.10]; plain -0.12 [-0.15 , -0.09]), and the tritone does not differ from them ($+0.03$ [-0.04 , $+0.09$]). This contrast was chosen after seeing the profile. The individual bonuses under M2 (plain means, point estimates) are $+0.14$, $+0.05$, $+0.18$, $+0.20$, $+0.16$, $+0.07$, $+0.19$, $+0.14$, $+0.15$, $+0.04$ and $+0.13$ from the minor second to the major seventh.

The step across analysis choices and spectra. Table S5 repeats the step under M2 with other kernel widths, exclusion windows and raw 0.25-semitone bins. The plain-mean step is stable across these choices (kernel SD 0.1: $+0.12$ [$+0.07$, $+0.17$]; 0.3: $+0.08$ [$+0.05$, $+0.11$]; window ± 0.4 : $+0.09$ [$+0.06$, $+0.13$]; ± 0.8 : $+0.10$ [$+0.07$, $+0.13$]; bins: $+0.13$ [$+0.07$, $+0.18$]), but in every variant it comes mainly from the guitar and piano, whose spectra are proxies from one analysed tone each, and it is absent for the synthetic harmonic tone and the flute.

Table S4: Decomposition of the held-out gap under three curves (exploratory). For each: plain mean over the six harmonic spectra [participant-bootstrap CI] and random effects [Hartung–Knapp CI]. M1: H&K + harmonicity (20 cents); M2: + interval size; M3: + sharpness proxy. Bold: CI excludes zero.

Quantity	M1		M2		M3	
	Plain	Random effects	Plain	Random effects	Plain	Random effects
Gap left, $r(8) - r(6)$	+0.19 [+0.13, +0.25]	+0.18 [+0.04, +0.32]	+0.22 [+0.16, +0.28]	+0.21 [+0.06, +0.36]	+0.21 [+0.16, +0.27]	+0.21 [+0.06, +0.35]
Bonus part, $b(8) - b(6)$	+0.07 [+0.01, +0.12]	+0.06 [-0.03, +0.16]	+0.07 [+0.01, +0.13]	+0.06 [-0.03, +0.16]	+0.07 [+0.01, +0.13]	+0.06 [-0.03, +0.16]
Baseline part, $q(8) - q(6)$	+0.12 [+0.09, +0.16]	+0.12 [-0.01, +0.25]	+0.15 [+0.11, +0.19]	+0.15 [+0.02, +0.28]	+0.15 [+0.11, +0.19]	+0.14 [+0.01, +0.27]
Step (indicator, 1–11)	+0.06 [+0.02, +0.10]	+0.06 [-0.06, +0.18]	+0.10 [+0.06, +0.14]	+0.10 [-0.03, +0.23]	+0.10 [+0.06, +0.13]	+0.10 [-0.03, +0.22]
Step (indicator, 1–14)	+0.11 [+0.07, +0.14]	+0.10 [+0.01, +0.20]	+0.13 [+0.10, +0.16]	+0.13 [+0.03, +0.23]	+0.12 [+0.09, +0.16]	+0.12 [+0.02, +0.22]
Step (7→8 increment)	+0.10 [+0.07, +0.12]	+0.10 [+0.01, +0.18]	+0.10 [+0.07, +0.13]	+0.10 [+0.01, +0.19]	+0.10 [+0.07, +0.13]	+0.10 [+0.01, +0.19]
Mean bonus, other intervals	+0.14 [+0.12, +0.15]	+0.13 [+0.08, +0.19]	+0.14 [+0.12, +0.15]	+0.14 [+0.08, +0.19]	+0.14 [+0.12, +0.15]	+0.14 [+0.08, +0.19]
Tritone bonus	+0.07 [+0.03, +0.11]	+0.06 [0.00, +0.13]	+0.07 [+0.03, +0.11]	+0.07 [0.00, +0.14]	+0.07 [+0.03, +0.11]	+0.07 [0.00, +0.14]
Other – tritone bonus	+0.07 [+0.03, +0.11]	+0.07 [0.00, +0.14]	+0.07 [+0.03, +0.11]	+0.07 [-0.01, +0.14]	+0.07 [+0.03, +0.11]	+0.07 [0.00, +0.14]
Tritone – M2/m7 bonus	+0.02 [-0.03, +0.07]	+0.02 [-0.05, +0.08]	+0.03 [-0.02, +0.08]	+0.03 [-0.04, +0.09]	+0.03 [-0.02, +0.07]	+0.02 [-0.04, +0.09]
M2/m7 – other bonus	-0.11 [-0.14, -0.08]	-0.11 [-0.13, -0.09]	-0.12 [-0.15, -0.09]	-0.12 [-0.14, -0.10]	-0.12 [-0.15, -0.09]	-0.12 [-0.14, -0.10]

Table S5: The step (M2) under other analysis choices (exploratory): plain mean over the six harmonic spectra [participant-bootstrap CI] of the step as an indicator and as an increment, the bonus and baseline parts of the gap, and the step per spectrum (point estimates). Guitar and piano use proxy spectra. Bold: CI excludes zero.

Variant	Six harmonic spectra				Step per spectrum					
	Step	Increment	Bonus part	Baseline part	Harm.	5 eq.	No 3rd	Flute	Guitar	Piano
Default (SD 0.2, ± 0.6)	+0.10 [+0.07, +0.13]	+0.10 [+0.07, +0.13]	+0.07 [+0.01, +0.12]	+0.15 [+0.12, +0.18]	-0.02	+0.04	+0.09	+0.01	+0.31	+0.17
Kernel SD 0.1	+0.12 [+0.07, +0.17]	+0.10 [+0.06, +0.14]	+0.09 [0.00, +0.19]	+0.14 [+0.10, +0.19]	+0.03	+0.03	+0.11	-0.01	+0.35	+0.22
Kernel SD 0.3	+0.08 [+0.05, +0.11]	+0.09 [+0.07, +0.12]	+0.04 [+0.01, +0.07]	+0.15 [+0.12, +0.18]	-0.03	+0.02	+0.07	+0.02	+0.27	+0.14
Exclusion ± 0.4	+0.09 [+0.06, +0.13]	+0.10 [+0.07, +0.13]	+0.07 [+0.01, +0.12]	+0.15 [+0.11, +0.18]	-0.03	+0.04	+0.08	0.00	+0.31	+0.16
Exclusion ± 0.8	+0.10 [+0.07, +0.13]	+0.10 [+0.07, +0.13]	+0.07 [+0.01, +0.12]	+0.15 [+0.12, +0.18]	-0.02	+0.04	+0.09	+0.01	+0.31	+0.17
Raw 0.25-semitone bins	+0.13 [+0.07, +0.18]	+0.11 [+0.07, +0.16]	+0.08 [-0.03, +0.18]	+0.15 [+0.09, +0.20]	+0.03	+0.03	+0.12	0.00	+0.34	+0.25

S5 Stretched and compressed partials

The dip at the stretched scale. Repeating the analysis at the stretched scale’s own fourth, tritone, fifth and minor sixth, the stretched tones show an observed dip of +0.19 [+0.07, +0.31] at 6.42 semitones (paired difference from the dip at 6, on the same bootstrap draws, +0.22 [+0.07, +0.37]). Calibrated H&K roughness fitted to the stretched experiment itself leaves +0.11 [-0.01, +0.23] of it, and adding harmonicity changes little (+0.11 [-0.02, +0.23]). The gap to the stretched minor sixth is +0.06 [-0.07, +0.20]. For compressed tones the dip at 5.56 is +0.07 [-0.05, +0.19] (paired difference +0.04 [-0.14, +0.22]). For harmonic tones the procedure returns the ordinary estimates (dip +0.38 [+0.26, +0.50]).

The bonus at the stretched scale (A4). If the in-tune bonus were attached to intervals as heard relative to the heard octave, it should appear at the stretched scale’s own intervals. Table S6 gives the mean bonus of the intervals other than the tritone at the scale matched to each spectrum and at the equal-tempered positions, with the minimum effect detectable with 80% power (2.8 SE). For US listeners the bonus is +0.13 [+0.10, +0.17] for harmonic tones and +0.02 [-0.02, +0.05] and +0.01 [-0.02, +0.05] at the stretched and compressed scales, with detectable effects of 0.05 and 0.05. The absence is therefore not a matter of power.

Timbre-relative harmonicity. The published harmonicity model cannot follow the shifted dip, because its template is a harmonic series. As an exploratory step taken after the screen, we replaced the template with the upper tone’s own partials and let the pitch-class circle wrap at the tone’s own octave, one way of reading the proposal that harmonic templates are learned from the sounds heard (Shamma & Klein, 2000). Out of sample, this model predicts a dip of +0.17 at 6.42 semitones for the stretched tones and leaves +0.02 [-0.11, +0.14], against +0.08 [-0.05, +0.20] with the standard template (paired difference -0.06 [-0.07, -0.05]); Marjeh et al.’s revised roughness also leaves little there (-0.01). With one stretched experiment neither residual is distinguishable from zero. The model pays for this on harmonic tones: the dip left

Table S6: Mean in-tune bonus of the intervals other than the tritone (exploratory; M2 curve) at the scale matched to each spectrum and at equal-tempered positions [participant-bootstrap CI], and the minimum detectable effect (MDE, 2.8 SE). Bold: CI excludes zero.

Spectrum (cohort)	Own scale	Equal-tempered	MDE
Harmonic (US)	+0.13 [+0.10 , +0.17]	–	0.05
Stretched 2.1 (US)	+0.02 [–0.02, +0.05]	–0.01 [–0.04, +0.02]	0.05
Compressed 1.9 (US)	+0.01 [–0.02, +0.05]	+0.03 [–0.01, +0.06]	0.05
Harmonic (Korea)	+0.07 [+0.03 , +0.12]	–	0.06
Stretched 2.1 (Korea)	–0.04 [–0.08, +0.01]	0.00 [–0.04, +0.04]	0.07
Compressed 1.9 (Korea)	–0.01 [–0.05, +0.04]	+0.04 [0.00, +0.09]	0.06

at 600 cents rises to +0.19 (standard template +0.14; paired difference +0.05 [+0.04, +0.06]) and the held-out R^2 falls to 0.515 (paired difference –0.018 [–0.037, –0.001]). Adding it to the 20-cent term does not improve the fit (paired R^2 difference –0.014 [–0.022, –0.008]). In summary, a template matched to the timbre can reproduce the stretched-scale dip but is not a better general curve, and a single stretched spectrum cannot decide between it and revised roughness.

S6 Roll-off

Table S7 gives the roll-off experiment at the three pre-set roll-off levels. The roughness curve predicts almost no gap at any roll-off, and the observed gap does not shrink as the upper partials fade.

Table S7: Study 3 of Marjeh et al. (2024) (harmonic tones, roll-off varied continuously; $N = 295$), smoothed at three roll-off levels (kernel SD 1.5 dB/octave). Pred.: gap predicted by the calibrated roughness curve. SD units [95% CI]. Bold: CI excludes zero.

dB/oct	Observed gap	Roughness pred.	After roughness	After rough. + harm.	Observed dip	Dip after rough. + harm.
2	+0.27 [+0.10 , +0.44]	+0.01 [+0.01, +0.02]	+0.26 [+0.09 , +0.43]	+0.20 [+0.04 , +0.37]	+0.41 [+0.26 , +0.57]	+0.20 [+0.04 , +0.34]
7	+0.18 [+0.05 , +0.32]	0.00 [–0.01, +0.01]	+0.18 [+0.05 , +0.32]	+0.13 [0.00 , +0.26]	+0.35 [+0.23 , +0.47]	+0.14 [+0.02 , +0.26]
12	+0.29 [+0.16 , +0.43]	0.00 [0.00, 0.00]	+0.29 [+0.16 , +0.43]	+0.25 [+0.12 , +0.38]	+0.36 [+0.21 , +0.50]	+0.22 [+0.08 , +0.36]

S7 Korean listeners

Korean listeners rating the same harmonic stimuli show a smaller gap than US listeners (difference –0.13 [–0.37, +0.10]; after roughness and harmonicity –0.14 [–0.39, +0.08]). Their dip at the tritone is clear (+0.25 [+0.10, +0.42]) and the two-mechanism curve accounts for it (–0.04). The cohort difference is not specific to harmonic tones: Korean listeners also rate the minor sixth lower relative to the tritone for stretched (–0.15 [–0.36, +0.07]) and compressed (–0.23 [–0.45, –0.02]) partials, where US listeners show no gap.

In-sample fits. Fitted to each Korean experiment alone (in-sample), roughness plus harmonicity at 20 cents reaches R^2 of 0.01–0.12; adding a linear term in interval size raises the mean to 0.87, and adding the sharpness proxy instead raises the range to 0.87–0.93, close to but below the split-half ceilings (0.946–0.950). These in-sample figures overstate what the models predict.

Held-out fits (A3). Table S8 predicts the Korean experiments with weights fitted to all US experiments. The recommended curve does worse than each experiment’s mean (mean held-out R^2 –0.19 [–0.26, –0.08]), as does the composite (–0.13 [–0.21, –0.03]); a linear term in

interval size brings the mean to +0.11 [−0.01, +0.22] and the sharpness proxy to +0.24 [+0.09, +0.36]. What the curves miss for Korean listeners is mainly a steeper decline of pleasantness with interval size. Sharpness is one candidate cause; in an online study, differences in playback equipment between cohorts are another.

Table S8: Korean experiments predicted with weights from US listeners (exploratory): held-out R^2 [participant-bootstrap CI], and the split-half ceiling.

Model	Harmonic	Stretched 2.1	Compressed 1.9	Mean
H&K + H (20)	+0.03 [−0.16, +0.23]	−0.15 [−0.28, −0.02]	−0.43 [−0.56, −0.25]	−0.19 [−0.26, −0.08]
+ interval size	+0.28 [+0.09, +0.43]	+0.15 [+0.01, +0.29]	−0.10 [−0.25, +0.08]	+0.11 [−0.01, +0.22]
+ sharpness proxy	+0.40 [+0.21, +0.55]	+0.37 [+0.18, +0.53]	−0.04 [−0.20, +0.14]	+0.24 [+0.09, +0.36]
Composite (Marjeh et al.)	+0.01 [−0.19, +0.19]	−0.11 [−0.20, 0.00]	−0.29 [−0.39, −0.13]	−0.13 [−0.21, −0.03]
Split-half ceiling	0.95	0.95	0.95	

S8 The smooth trend and the sharpness proxy

All analyses in this section are exploratory, except the triad test, whose predictions were written down before the triad data were read.

The proxy. S is the loudness-weighted mean critical-band rate of the partials (weights amplitude^{0.6}) with Zwicker’s high-frequency weighting, $g(z) = 1$ below 15.8 Bark and $0.066 e^{0.171z}$ above (Fastl & Zwicker, 2007). It is computed from the synthesis parameters, not from audio, and is not a calibrated sharpness in acum. It was chosen over interval size and the log spectral centroid by the same held-out criterion that we report, so its advantage over them is optimistic.

Pooled weights. Pooling all ten US experiments, the three-term curve with the 20-cent harmonicity tolerance has weights $b_R = -1.84$, $b_H = 0.68$ and $b_S = -0.065$ per unit of S (weighted Bark), in the units of the two-term curve of main-text Section 6.1. We do not propose it as the recommended curve, because the proxy was chosen after seeing the residuals and failed its pre-specified test (Section S8).

The roll-off check. Fitted separately at five roll-offs (kernel SD 1.5 dB/octave), the slope over interval size of the residual after roughness and harmonicity flattens as the roll-off steepens (Table S9; difference between 1 and 13 dB/octave -0.33 [−0.49, −0.17]). With one weight shared across roll-offs, sharpness fits better than interval size (mean R^2 0.74 against 0.70; difference +0.040 [+0.022, +0.053]; roughness and harmonicity alone 0.49), but the slopes it implies cover only about 50% of the observed flattening. The two terms are strongly correlated, so the sign of the interval-size term with both in the model (0.33 [0.15, 0.54] per octave; sharpness -0.20 [−0.27, −0.15]) is weak evidence. We looked at this experiment before choosing sharpness, so it motivated the hypothesis as much as it tests it.

The pre-specified triad test. Before reading them, we wrote down (but did not publicly register) three predictions for the one rating set of Marjeh et al. not yet analysed: 80 triads of five equal harmonics with and without the third harmonic, rated by 192 participants. Fitted to the triads, the sharpness weight should be negative; adding sharpness should improve leave-one-chord-out R^2 ; and the three-term curve with frozen dyad weights should correlate better with

Table S9: Roll-off experiment (exploratory): slope over interval size of the residual after H&K roughness plus harmonicity (20 cents), SD per octave of interval [participant-bootstrap CI], and the slope implied by the sharpness proxy with one shared weight.

Roll-off (dB/octave)	Observed slope	Implied by sharpness
1	-0.45 [-0.60, -0.33]	-0.41
4	-0.41 [-0.51, -0.30]	-0.37
7	-0.39 [-0.48, -0.30]	-0.32
10	-0.29 [-0.40, -0.20]	-0.28
13	-0.13 [-0.22, -0.03]	-0.25

the chord means than the two-term curve. We also recorded the risk that the trend had been absent for five equal harmonics in the dyads. None of the three predictions passed. The fitted weight was negative but its interval included zero (-0.095 [-0.193 , $+0.010$]); leave-one-chord-out R^2 went from 0.748 to 0.753 ($+0.005$ [-0.009 , $+0.041$]); and the frozen-weight correlation went from 0.874 to 0.880 ($+0.006$ [-0.006 , $+0.017$]). The triads leave little room for a third term, but they do not confirm it.

S9 Equal temperament and just intonation (A8)

This analysis is exploratory; it was added after the dense-dyad residuals had been seen. It asks whether evaluating the gap at equal-tempered positions creates part of it. With a kernel of SD 0.1 semitones (sensitivity: 0.05 and 0.2), we located, per experiment with a participant-bootstrap CI, the listeners’ minor-sixth peak (argmax over 7.5–8.5 semitones), tritone trough (argmin over 5.5–6.5) and, as controls, the peaks of the fifth, major third and major sixth, on the listener profile and on the held-out predictions of each model (Table S10, Figure S2). We then recomputed the gap at equal-tempered positions and at two just-intoned pairs: 8/5 against 7/5 (JIa) and 8/5 against 45/32 (JIb) (Table S11).

Pooled over the six harmonic spectra, the listeners’ minor-sixth peak lies at 808 [796, 819] cents, -6 [-17 , $+6$] cents from the peak of the harmonicity model at the published tolerance (813 [812, 815]); at the narrower width it is 805 [797, 813] (difference -8 [-16 , 0]). The fifth peaks where the model does (-3 [-6 , $+1$]), and the major third and major sixth lie $+8$ [0 , $+17$] and $+5$ [-2 , $+12$] cents from the model’s peaks, each displaced from the model’s just-intoned peak towards equal temperament, as the minor sixth is, but none of these differences is established. The tritone region is a broad trough (616 [576, 655]) with no distinct minimum at 600 cents or at either just-intoned tritone. The gap left by the recommended curve is $+0.20$ [$+0.02$, $+0.38$] at equal-tempered positions and $+0.22$ [$+0.06$, $+0.37$] and $+0.19$ [$+0.05$, $+0.33$] at the just-intoned pairs; the paired differences, $+0.03$ [-0.11 , $+0.17$] and $+0.01$ [-0.10 , $+0.12$], show neither a smaller gap at just intonation nor equivalence within ± 0.05 SD. The result is inconclusive. The recommended curve has no interior minor-sixth or major-third peak at this width (rows marked “edge”), so it smooths away a peak that listeners show.

S10 Chords: the three-component decomposition

Table S12 and Figure S3 give the unique variance of interference (H&K), harmonicity and familiarity, each over the other two, in the Bowling and Johnson-Laird chords under four covariate specifications. With chord size controlled and equal-tempered scoring, the unique variance of harmonicity in the Bowling chords (0.037) was about half that of interference (0.075) and familiarity (0.080). Scored at the just intonation actually presented, the three are comparable

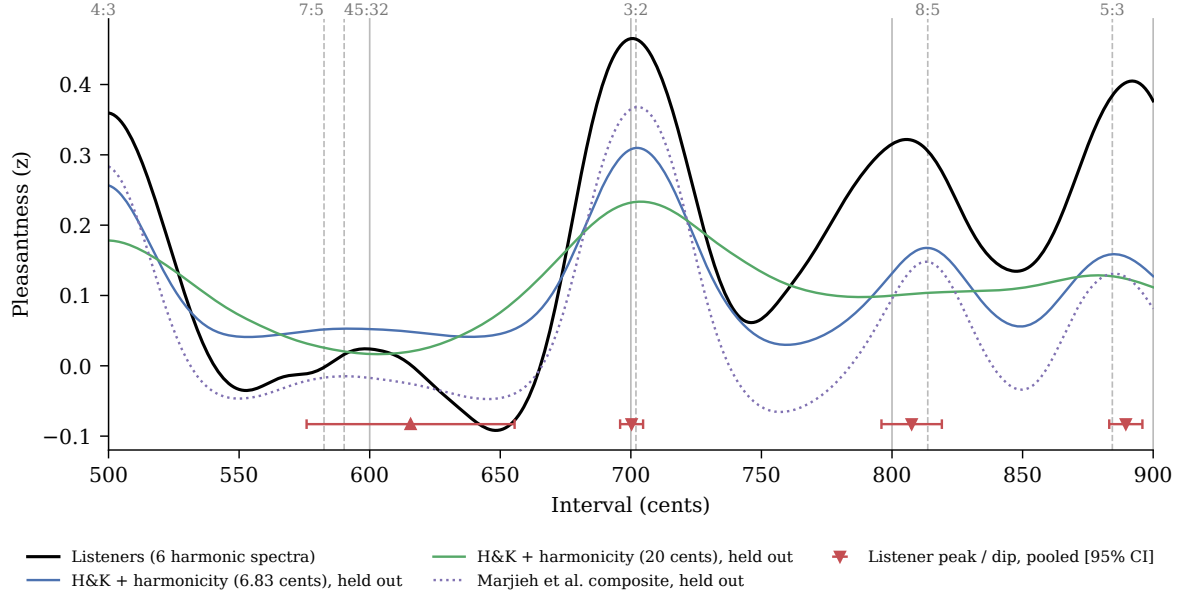


Figure S2: Pooled listener profile of the six harmonic spectra at kernel SD 0.1 semitones (black) with the held-out model curves, from 500 to 900 cents (exploratory). Solid grey lines: equal-tempered positions; dashed: just-intoned ratios. Red markers: pooled listener peak and trough locations [Hartung–Knapp CI].

(0.069, 0.068 and 0.061), and the full model’s R^2 rises from 0.738 to 0.775. Without a chord-size control familiarity appears to dominate (0.147 against 0.049 for interference), which is how an earlier analysis by the author reported it. Across the 298 chords, harmonicity scored in the two tunings correlates $r = 0.941$. The chord bootstraps resample chords, not raters.

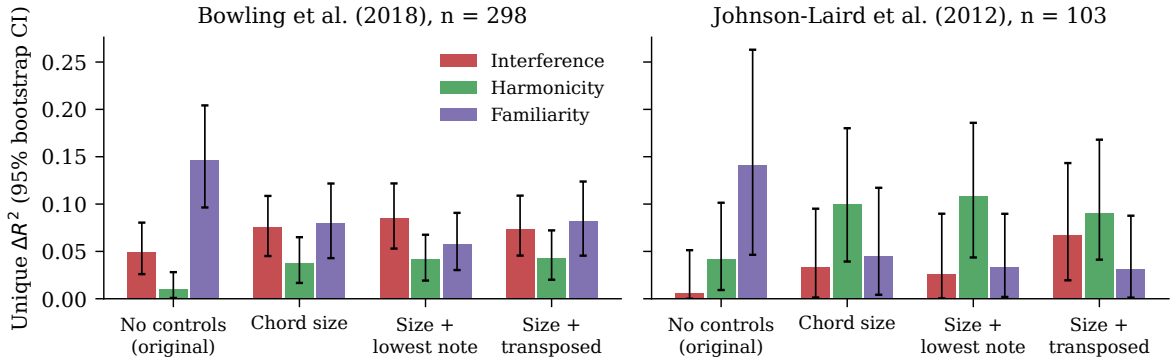


Figure S3: Unique ΔR^2 of interference, harmonicity and familiarity (95% bootstrap CI) under four covariate specifications, models scored in equal temperament.

S11 Roughness curves on measured spectra

Could the problem lie in idealised spectra? A dissonance curve computed from a measured spectrum might place the tritone above the minor sixth. We computed the tritone/minor-sixth roughness difference, $(D_{600} - D_{800})/D_{800}$, for 14 measured spectra under four roughness kernels (Sethares with product and minimum weighting, Vassilakis and H&K). The answer depends on the analysis more than on anything a listener would recognise (Table S13, Figure S4). On one

Philharmonia clarinet spectrum the four kernels give -46.6 , -3.4 , $+6.8$ and $+39.8\%$, a range of 86.4 pp. Across the eight C4 instruments the instrument accounts for 55% of the variance of the difference, the kernel for 8% and their interaction for 36% (three stable kernels). The shape constants of the H&K kernel, which are rarely reported, move the difference by a median 13.9 pp over a $\pm 20\%$ grid, and the recording (two clarinet libraries) by a median 6.0 pp (Table S15, Figure S5).

One ordering is robust: under H&K the clarinet rates the minor sixth rougher than the tritone at three registers in both libraries (Iowa: -22.1 , -43.0 and -11.7% ; Table S14, Figure S6). The reversal comes from a single pair of odd harmonics at the minor sixth (the fifth of the lower note against the third of the upper, 62 Hz apart, 70% of cross-tone roughness) and from the steep decay of the H&K kernel beyond its peak (Figure S7). Depending on instrument and kernel, roughness computed from real spectra can reverse the listeners' ordering as easily as reproduce it.

Extraction. Sustained forte notes were taken from the Philharmonia library (Philharmonia Orchestra, n.d.): cello, clarinet, oboe, flute, bassoon, saxophone and French horn at C4, plus low and high registers for clarinet (D3, C5), cello (C3, C5) and oboe (A#3, C5); a piano C4 and three further clarinet notes (D3, C4, C5) came from the Iowa library (University of Iowa Electronic Music Studios, n.d.). Each note was analysed with an 8192-point STFT (75% overlap) over the stable middle 60% of the sample. Peaks with prominence above 0.005 of the loudest partial were accepted and the 15 strongest kept; frequencies were refined by parabolic interpolation on log magnitude. The Iowa piano's partials follow the stiff-string law $f_n = nf_0\sqrt{1+Bn^2}$ (Fletcher, 1964) with $B = 3.17 \times 10^{-4}$ (SE 6.3×10^{-7}).

Kernels. Sethares (product): $e^{-3.5s\Delta f} - e^{-5.75s\Delta f}$ with $s = 0.24/(0.0207f_{\min} + 18.96)$, weighted by a_1a_2 (Sethares, 1993; Schlecht, 2020); Sethares (min): weighted by $\min(a_1, a_2)$ (Sethares, 2005); Vassilakis: weighted by $(a_1a_2)^{0.1}(2\min(a_1, a_2)/(a_1 + a_2))^{3.11}$ (Vassilakis, 2001); H&K as in main-text Section 3; as a sensitivity check the ERB of Glasberg and Moore (1990) replaced the critical bandwidth. Coincident partials were merged in quadrature. Each spectrum's dyad was swept from unison to the octave in 5-cent steps. Implementations differ between packages (Mashinter, 2006).

Sensitivity factors. Instrument (eight at C4); kernel; H&K constants on a wide grid, $a = 0.15, 0.2, 0.25, 0.3, 0.35$ crossed with $b = 1.0, 1.5, 2.0, 2.5, 3.0$, and on a narrow grid, $a = 0.2, 0.25, 0.3$ crossed with $b = 1.5, 2, 2.5$; recording (Philharmonia vs Iowa clarinet); extraction constants (prominence 0.005–0.0002, cap 15 or 25); audio codec (lossless to 48 kbit/s MP3); idealised-spectrum constants (5, 7 or 11 harmonics, roll-off 0.8–1.0). Under the narrow grid the sign of the H&K difference is stable for 11 of 17 spectra. Vassilakis's near-flat amplitude weight makes it depend on how many weak partials are admitted (extraction moves it by up to 168.3 pp; Table S18).

Clarinet. Table S16 tests the H&K clarinet reversal against every constant. It survives the extraction sweep (range at most 1.5 pp), the cutoff sweep, ERB substitution and the whole narrow a, b grid on both libraries (Table S17). At the minor sixth the fifth harmonic of the lower note (1304 Hz) and the third of the upper (1242 Hz) lie near the peak of the H&K kernel; at the tritone the fundamentals are 1.49 critical bandwidths apart, where the H&K weight is 0.17% of its peak but the Sethares kernel still assigns substantial roughness (36% of its cross-tone total). Listener ratings of clarinet dyads at several registers would discriminate between kernels.

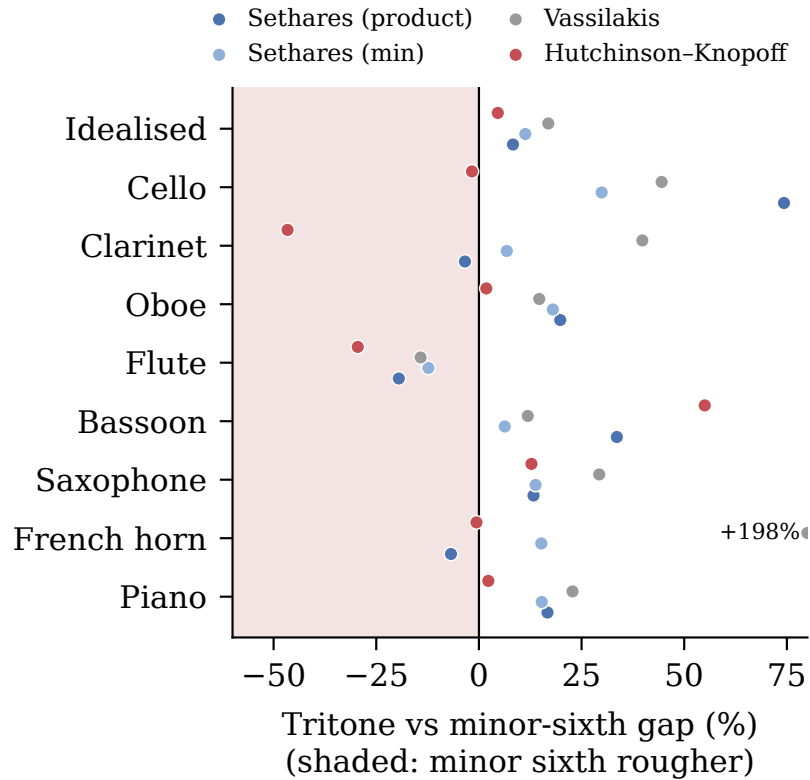


Figure S4: Tritone/minor-sixth roughness difference (%) per spectrum and kernel at C4 (shaded: the kernel rates the minor sixth rougher).

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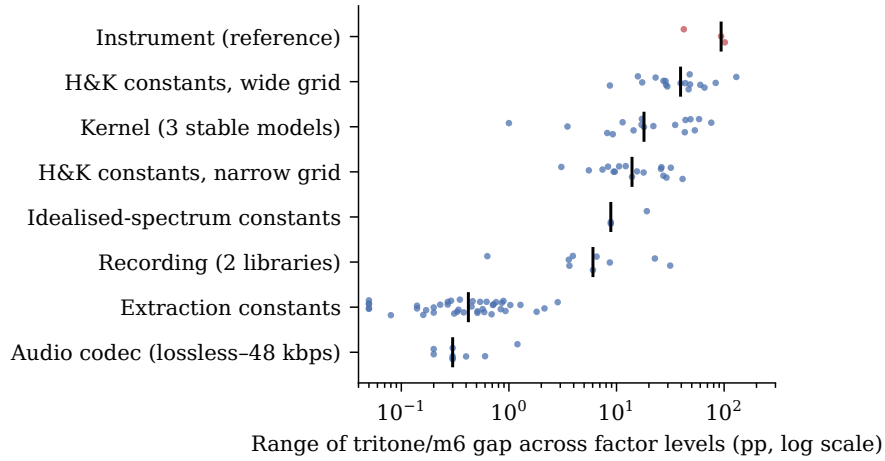


Figure S5: Range of the tritone/minor-sixth difference (pp, log scale) across the levels of each analysis factor, stable kernels; bars, medians. Red: instrument (reference).

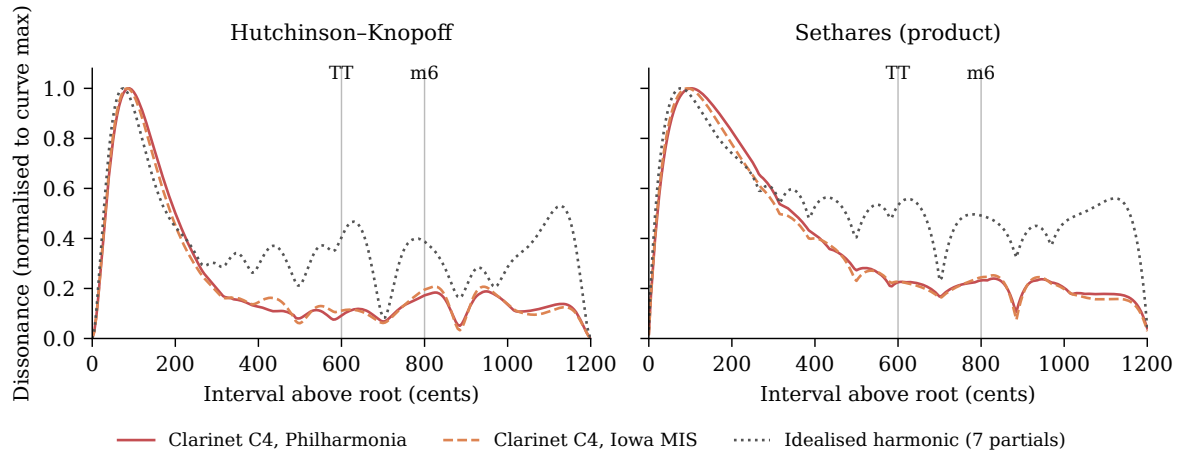


Figure S6: Dissonance curves (normalised to each curve's maximum) for two clarinet C4 recordings and an idealised seven-partial harmonic tone, under the H&K and Sethares (product) kernels.

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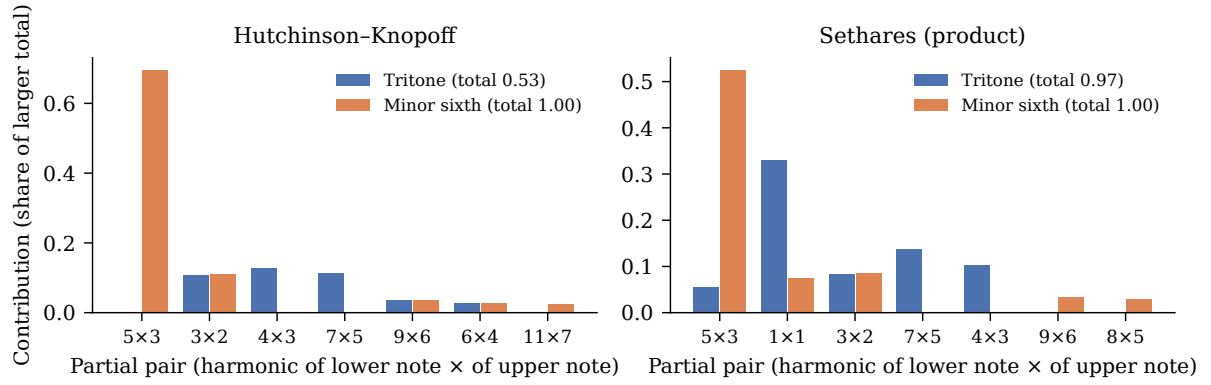


Figure S7: Philharmonia clarinet C4: the partial pairs contributing most roughness at the tritone and the minor sixth under each kernel (share of the larger total).

Table S10: Peak and trough locations (cents) [participant-bootstrap CI; pooled rows, random effects with Hartung–Knapp CI], kernel SD 0.1 semitones (exploratory). Equal-tempered and just-intoned positions: m6 800/814; TT 600/583 (7/5) or 590 (45/32); P5 700/702; M3 400/386; M6 900/884. [†]Point estimate on the edge of the search window. Edge: extremum on the window edge in at least one spectrum.

	m6 peak	TT trough	P5 peak	M3 peak	M6 peak
<i>Listeners</i>					
Harmonic, 3 dB/oct	806 [777, 825]	579 [550, 650]	700 [691, 708]	404 [378, 413]	894 [886, 900]
5 equal harmonics	815 [808, 824]	633 [550, 650]	696 [689, 715]	394 [388, 399]	880 [876, 897]
5 harmonics, no 3rd	850 [788, 850] [†]	552 [550, 650]	693 [686, 708]	387 [380, 398]	899 [857, 907]
Flute	801 [773, 813]	603 [554, 650]	702 [694, 708]	417 [379, 423]	898 [861, 906]
Guitar	810 [781, 818]	650 [562, 650] [†]	708 [699, 719]	392 [382, 401]	889 [881, 894]
Piano	799 [789, 809]	644 [634, 648]	700 [693, 706]	406 [382, 411]	888 [880, 898]
Pure tones	806 [750, 850]	633 [565, 641]	705 [691, 737]	401 [360, 450]	887 [866, 894]
Stretched (2.1)	750 [750, 755] [†]	650 [560, 650] [†]	744 [729, 750]	408 [355, 449]	941 [856, 950]
Compressed (1.9)	824 [750, 850]	550 [550, 627] [†]	650 [650, 750] [†]	377 [359, 416]	850 [850, 938] [†]
Pooled, 6 harmonic (RE)	808 [796, 819]	616 [576, 655]	700 [696, 705]	396 [387, 404]	890 [883, 896]
<i>H&K + harmonicity (6.83 cents)</i>					
Harmonic, 3 dB/oct	814 [814, 815]	644 [643, 645]	703 [702, 704]	387 [387, 388]	886 [885, 887]
5 equal harmonics	814 [813, 815]	588 [586, 589]	701 [701, 702]	385 [384, 386]	885 [884, 886]
5 harmonics, no 3rd	812 [812, 813]	550 [550, 550] [†]	701 [700, 701]	387 [387, 388]	881 [881, 882]
Flute	815 [814, 815]	632 [631, 633]	703 [702, 704]	388 [388, 389]	887 [886, 887]
Guitar	812 [812, 813]	650 [649, 650] [†]	703 [702, 704]	390 [389, 390]	882 [881, 883]
Piano	812 [812, 813]	649 [648, 650]	704 [703, 704]	389 [388, 390]	886 [885, 887]
Pure tones	813 [812, 814]	600 [600, 601]	702 [701, 702]	390 [389, 391]	950 [950, 950] [†]
Stretched (2.1)	751 [750, 752]	650 [650, 650] [†]	750 [750, 750] [†]	407 [406, 409]	949 [948, 950]
Compressed (1.9)	813 [812, 814]	595 [594, 596]	650 [650, 650] [†]	450 [390, 450] [†]	890 [889, 891]
Pooled, 6 harmonic (RE)	813 [812, 815]	edge	702 [701, 704]	388 [386, 389]	885 [882, 887]
<i>H&K + harmonicity (20 cents)</i>					
Harmonic, 3 dB/oct	850 [850, 850] [†]	624 [621, 627]	703 [703, 704]	387 [386, 388]	885 [884, 886]
5 equal harmonics	850 [850, 850] [†]	593 [592, 594]	702 [702, 703]	381 [381, 382]	883 [882, 883]
5 harmonics, no 3rd	810 [809, 810]	550 [550, 550] [†]	750 [750, 750] [†]	387 [387, 388]	850 [850, 850] [†]
Flute	750 [750, 750] [†]	610 [609, 612]	703 [702, 704]	450 [450, 450] [†]	888 [887, 888]
Guitar	750 [750, 750] [†]	601 [601, 602]	706 [705, 707]	450 [450, 450] [†]	850 [850, 850] [†]
Piano	750 [750, 750] [†]	605 [604, 606]	707 [706, 707]	450 [450, 450] [†]	850 [850, 850] [†]
Pure tones	750 [750, 750] [†]	600 [600, 601]	701 [700, 701]	450 [450, 450] [†]	950 [950, 950] [†]
Stretched (2.1)	753 [752, 753]	650 [650, 650] [†]	750 [750, 750] [†]	411 [411, 412]	947 [946, 948]
Compressed (1.9)	818 [817, 818]	586 [585, 587]	651 [650, 651]	450 [450, 450] [†]	905 [850, 906]
Pooled, 6 harmonic (RE)	edge	edge	edge	edge	edge
<i>Composite</i>					
Harmonic, 3 dB/oct	814 [813, 814]	649 [648, 650]	703 [702, 704]	387 [386, 388]	886 [885, 887]
5 equal harmonics	813 [813, 814]	592 [591, 592]	701 [701, 702]	386 [385, 387]	885 [884, 885]
5 harmonics, no 3rd	812 [812, 813]	550 [550, 550] [†]	701 [700, 701]	387 [387, 388]	882 [882, 883]
Flute	814 [813, 815]	635 [634, 635]	703 [702, 704]	388 [387, 389]	886 [885, 887]
Guitar	813 [812, 813]	648 [648, 649]	703 [702, 704]	388 [388, 389]	883 [882, 950]
Piano	813 [812, 813]	649 [648, 649]	704 [703, 705]	388 [387, 389]	886 [885, 887]
Pure tones	813 [812, 814]	600 [600, 601]	702 [701, 702]	389 [388, 390]	950 [950, 950] [†]
Stretched (2.1)	751 [750, 753]	650 [649, 650] [†]	750 [750, 750] [†]	410 [408, 412]	948 [947, 949]
Compressed (1.9)	813 [812, 814]	599 [598, 600]	650 [650, 650] [†]	450 [450, 450] [†]	942 [892, 944]
Pooled, 6 harmonic (RE)	813 [812, 814]	edge	702 [701, 704]	387 [386, 388]	885 [883, 887]

Table S11: Gap at equal-tempered (ET) and just-intoned positions (J1a: 8/5 vs 7/5; J1b: 8/5 vs 45/32), observed and left by each held-out model, random effects over the six harmonic spectra [Hartung–Knapp CI], at three kernel widths (semitones; exploratory). A kernel of 0.2 semitones cannot separate positions 14 cents apart.

SD		ET	J1a	J1b	J1a – ET	J1b – ET
0.1	Observed	+0.28 [+0.13, +0.43]	+0.29 [+0.18, +0.40]	+0.27 [+0.17, +0.36]	+0.03 [−0.12, +0.17]	+0.01 [−0.11, +0.12]
	Left by H&K	+0.28 [+0.13, +0.43]	+0.29 [+0.17, +0.41]	+0.27 [+0.16, +0.37]	+0.02 [−0.12, +0.17]	0.00 [−0.11, +0.12]
	Left by H&K + H (6.83)	+0.20 [+0.02, +0.38]	+0.18 [+0.02, +0.34]	+0.16 [+0.02, +0.30]	−0.01 [−0.15, +0.13]	−0.03 [−0.14, +0.08]
	Left by H&K + H (20)	+0.20 [+0.02, +0.38]	+0.22 [+0.06, +0.37]	+0.19 [+0.05, +0.33]	+0.03 [−0.11, +0.17]	+0.01 [−0.10, +0.12]
	Left by composite	+0.17 [−0.03, +0.37]	+0.13 [−0.05, +0.31]	+0.11 [−0.05, +0.28]	−0.03 [−0.17, +0.12]	−0.04 [−0.15, +0.07]
0.05	Observed	+0.32 [+0.08, +0.56]	+0.35 [+0.20, +0.50]	+0.26 [+0.16, +0.36]	+0.05 [−0.25, +0.34]	−0.04 [−0.29, +0.21]
	Left by H&K	+0.32 [+0.08, +0.57]	+0.34 [+0.19, +0.50]	+0.25 [+0.14, +0.36]	+0.04 [−0.25, +0.33]	−0.05 [−0.29, +0.20]
	Left by H&K + H (6.83)	+0.24 [−0.04, +0.51]	+0.21 [+0.02, +0.39]	+0.11 [−0.03, +0.25]	0.00 [−0.29, +0.28]	−0.09 [−0.33, +0.15]
	Left by H&K + H (20)	+0.24 [−0.04, +0.51]	+0.26 [+0.09, +0.44]	+0.16 [+0.02, +0.29]	+0.05 [−0.23, +0.34]	−0.04 [−0.28, +0.20]
	Left by composite	+0.20 [−0.10, +0.50]	+0.15 [−0.05, +0.35]	+0.06 [−0.11, +0.22]	−0.03 [−0.31, +0.26]	−0.11 [−0.34, +0.13]
0.2	Observed	+0.26 [+0.15, +0.36]	+0.25 [+0.16, +0.34]	+0.25 [+0.16, +0.33]	0.00 [−0.05, +0.04]	−0.01 [−0.04, +0.03]
	Left by H&K	+0.26 [+0.14, +0.37]	+0.25 [+0.15, +0.35]	+0.24 [+0.15, +0.34]	−0.01 [−0.05, +0.04]	−0.01 [−0.05, +0.02]
	Left by H&K + H (6.83)	+0.19 [+0.06, +0.33]	+0.18 [+0.05, +0.30]	+0.17 [+0.05, +0.29]	−0.02 [−0.06, +0.02]	−0.02 [−0.05, +0.01]
	Left by H&K + H (20)	+0.18 [+0.04, +0.32]	+0.18 [+0.05, +0.32]	+0.17 [+0.04, +0.30]	0.00 [−0.04, +0.04]	−0.01 [−0.04, +0.02]
	Left by composite	+0.17 [+0.02, +0.32]	+0.15 [+0.01, +0.29]	+0.14 [+0.01, +0.28]	−0.02 [−0.06, +0.02]	−0.02 [−0.05, +0.01]

Table S12: Unique ΔR^2 [95% bootstrap CI] (p , nested F) of interference (H&K), harmonicity and familiarity over the other two, under four covariate specifications. R^2 : full model including covariates.

Covariates	R^2	Interference	Harmonicity	Familiarity
<i>Bowling et al. (2018), $n = 298$</i>				
None (as originally reported)	0.674	0.049 [0.026, 0.080] (<.001)	0.010 [0.001, 0.028] (.004)	0.147 [0.096, 0.204] (<.001)
Chord size	0.738	0.075 [0.045, 0.109] (<.001)	0.037 [0.017, 0.065] (<.001)	0.080 [0.043, 0.122] (<.001)
Chord size + lowest note	0.749	0.085 [0.053, 0.122] (<.001)	0.042 [0.019, 0.068] (<.001)	0.058 [0.030, 0.091] (<.001)
Chord size; models on transposed chords	0.737	0.074 [0.046, 0.109] (<.001)	0.043 [0.020, 0.072] (<.001)	0.082 [0.045, 0.124] (<.001)
<i>Bowling et al. (2018), interference and harmonicity rescored at the presented just-intonation pitches</i>				
None (as originally reported)	0.692	0.038 [0.017, 0.065] (<.001)	0.024 [0.008, 0.050] (<.001)	0.133 [0.086, 0.184] (<.001)
Chord size	0.775	0.069 [0.040, 0.102] (<.001)	0.068 [0.041, 0.101] (<.001)	0.061 [0.032, 0.095] (<.001)
Chord size + lowest note	0.786	0.080 [0.048, 0.117] (<.001)	0.074 [0.045, 0.104] (<.001)	0.043 [0.021, 0.068] (<.001)
Chord size; models on transposed chords	0.775	0.069 [0.042, 0.104] (<.001)	0.071 [0.042, 0.104] (<.001)	0.064 [0.035, 0.097] (<.001)
<i>Johnson-Laird et al. (2012), $n = 103$</i>				
None (as originally reported)	0.414	0.005 [0.000, 0.051] (.344)	0.042 [0.009, 0.101] (.009)	0.141 [0.046, 0.263] (<.001)
Chord size	0.500	0.033 [0.001, 0.095] (.013)	0.100 [0.039, 0.180] (<.001)	0.045 [0.004, 0.117] (.004)
Chord size + lowest note	0.516	0.026 [0.000, 0.090] (.025)	0.108 [0.044, 0.186] (<.001)	0.033 [0.002, 0.090] (.011)
Chord size; models on transposed chords	0.534	0.068 [0.019, 0.143] (<.001)	0.090 [0.041, 0.168] (<.001)	0.031 [0.001, 0.088] (.012)

Table S13: Tritone/minor-sixth roughness difference (%) at C4. Bold: negative (the kernel rates the minor sixth rougher). Vassilakis values are highly sensitive to extraction settings and should be read for sign only.

Spectrum	Sethares	Sethares-min	Vassilakis	H&K
Idealised (7 harmonics, 0.9^{n-1})	+8.3	+11.3	+16.9	+4.6
Cello	+74.3	+29.9	+44.5	−1.7
Clarinet	−3.4	+6.8	+39.8	−46.6
Oboe	+19.8	+18.0	+14.7	+1.8
Flute	−19.5	−12.3	−14.2	−29.5
Bassoon	+33.6	+6.3	+11.9	+55.0
Saxophone	+13.3	+13.8	+29.3	+12.8
French horn	−6.8	+15.2	+198.1	−0.6
Piano (Iowa)	+16.7	+15.3	+22.8	+2.3
Negative gaps (of 8 measured)	3	1	1	4

Table S14: Tritone/minor-sixth roughness difference (%) at three registers (Philharmonia). Bold: negative.

Spectrum	Sethares	Sethares-min	Vassilakis	H&K
Clarinet D3	+12.3	+11.9	+21.8	− 30.8
Clarinet C4	− 3.4	+6.8	+39.8	− 46.6
Clarinet C5	+15.3	+4.0	+65.9	− 43.2
Cello C3	+7.6	+11.1	+13.8	+9.5
Cello C4	+74.3	+29.9	+44.5	− 1.7
Cello C5	− 6.9	+1.3	+46.9	− 5.0
Oboe A $\sharp$ 3	− 9.1	− 4.5	+8.3	− 21.6
Oboe C4	+19.8	+18.0	+14.7	+1.8
Oboe C5	+18.7	+23.1	+179.7	+11.7

Table S15: Range of the tritone/minor-sixth roughness difference (pp) across the levels of each analysis factor. n : number of (spectrum, kernel) cells for the three stable kernels. For the instrument row each cell is one kernel across the eight C4 instruments. † H&K only. ‡ For Vassilakis this row shows its amplitude exponent varied over $\{0.1, 0.25, 0.5, 1\}$.

Factor	n	Stable kernels		Vassilakis	
		Median	Max	Median	Max
Instrument (8 at C4; reference quantity)	3	93.8	101.6	212.3	212.3
Kernel (Sethares, Sethares-min, H&K)	17	18.0	76.0	48.7	204.9
H&K constants a, b , wide grid †	17	39.4	129.8	24.5	473.9 ‡
H&K constants a, b , narrow grid †	17	13.9	41.1	—	—
Idealised-spectrum constants (partials, roll-off)	3	8.8	19.1	24.6	24.6
Recording (Philharmonia vs Iowa clarinet)	9	6.0	31.5	23.6	51.7
Extraction constants (prominence, cap)	42	0.4	2.8	9.3	168.3
Audio codec (lossless to 48 kbit/s)	9	0.3	1.2	4.3	5.8

Table S16: Robustness of the H&K clarinet difference (%). Extraction: range (pp) across the extraction sweep. Cutoff: range with the 1.2-CBW cutoff at 1.2, 1.5, 2.0 or removed (Philharmonia only). Narrow/wide: range over the a, b grids. ERB: difference with the ERB in place of the CBW. Bold: positive.

Reg.	Library	Diff.	Extraction	Cutoff	Narrow a, b	Wide a, b	ERB
D3	Philharmonia	−30.8	0.54	[−30.8, −29.0]	[−40.6, −13.3]	[−45.1, + 1.6]	−36.5
D3	Iowa	−22.1	1.46	—	[−32.7, −6.5]	[−35.3, + 4.1]	−26.2
C4	Philharmonia	−46.6	0.27	[−46.6, −45.6]	[−50.7, −38.5]	[−54.0, −10.4]	−46.5
C4	Iowa	−43.0	0.03	—	[−47.1, −33.2]	[−49.6, −19.9]	−46.6
C5	Philharmonia	−43.2	0.33	[−43.2, −42.5]	[−45.2, −39.7]	[−47.1, −19.8]	−46.3
C5	Iowa	−11.7	0.03	—	[−13.2, −10.2]	[−14.0, −5.3]	−15.7

Table S17: H&K tritone/minor-sixth difference (%) per spectrum under the a, b grids and ERB substitution. Bold: sign differs from the baseline.

Spectrum	Baseline	Narrow a, b	Wide a, b	ERB, cut 1.2	ERB, no cut
Bassoon C4 (Phil.)	+55.0	[+43.3, +58.8]	[+14.0, +61.9]	+54.2	+51.9
Cello C3 (Phil.)	+9.5	[−3.9, +14.0]	[−14.0, +14.8]	+12.1	+12.7
Cello C4 (Phil.)	−1.7	[−30.1, +11.0]	[−55.3, +28.2]	+7.0	+5.8
Cello C5 (Phil.)	−5.0	[−9.7, −2.3]	[−16.4, −0.6]	−7.9	−8.0
Clarinet C4 (Phil.)	−46.6	[−50.7, −38.5]	[−54.0, −10.4]	−46.5	−45.6
Clarinet C5 (Phil.)	−43.2	[−45.2, −39.7]	[−47.1, −19.8]	−46.3	−46.0
Clarinet D3 (Phil.)	−30.8	[−40.6, −13.3]	[−45.1, +1.6]	−36.5	−35.2
Flute C4 (Phil.)	−29.5	[−33.0, −22.4]	[−35.8, −12.7]	−33.5	−33.1
French horn C4 (Phil.)	−0.6	[−9.4, +0.2]	[−27.5, +1.0]	−0.7	−1.1
Oboe A#3 (Phil.)	−21.6	[−25.1, −15.7]	[−27.2, +21.1]	−20.8	−20.0
Oboe C4 (Phil.)	+1.8	[−10.3, +21.6]	[−17.1, +112.8]	−0.2	+0.3
Oboe C5 (Phil.)	+11.7	[−6.0, +23.2]	[−30.1, +30.0]	+10.2	+10.1
Saxophone C4 (Phil.)	+12.8	[+6.6, +15.0]	[−0.8, +16.5]	+13.0	+12.6
Piano C4 (Iowa)	+2.3	[−16.9, +9.0]	[−39.5, +26.3]	+4.2	+3.4
Clarinet D3 (Iowa)	−22.1	[−32.7, −6.5]	[−35.3, +4.1]	−26.2	−25.3
Clarinet C4 (Iowa)	−43.0	[−47.1, −33.2]	[−49.6, −19.9]	−46.6	−45.8
Clarinet C5 (Iowa)	−11.7	[−13.2, −10.2]	[−14.0, −5.3]	−15.7	−15.7

Table S18: Vassilakis tritone/minor-sixth difference (%) as the amplitude exponent varies (published value 0.1). Bold: sign differs from the published-exponent value.

Spectrum	0.1	0.25	0.5	1.0
Bassoon C4 (Phil.)	+11.9	+4.6	−3.9	+8.6
Cello C3 (Phil.)	+13.8	+16.5	+18.6	+16.1
Cello C4 (Phil.)	+44.5	+49.5	+76.0	+228.8
Cello C5 (Phil.)	+46.9	+34.2	+10.2	−4.0
Clarinet C4 (Phil.)	+39.8	+38.5	+37.5	+65.8
Clarinet C5 (Phil.)	+65.9	+69.9	+94.5	+539.9
Clarinet D3 (Phil.)	+21.8	+24.6	+32.7	+59.7
Flute C4 (Phil.)	−14.2	−18.9	−26.2	−37.5
French horn C4 (Phil.)	+198.1	+182.2	+175.5	+200.0
Oboe A#3 (Phil.)	+8.3	−2.0	−14.8	−23.4
Oboe C4 (Phil.)	+14.7	+14.7	+17.0	+36.7
Oboe C5 (Phil.)	+179.7	+102.2	+32.0	+3.1
Saxophone C4 (Phil.)	+29.3	+28.4	+28.3	+36.8
Piano C4 (Iowa)	+22.8	+23.6	+31.3	+84.2
Clarinet D3 (Iowa)	+3.4	+2.9	+3.8	+14.3
Clarinet C4 (Iowa)	+16.2	+8.9	+3.7	+27.6
Clarinet C5 (Iowa)	+14.2	+9.7	+6.1	+12.9